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No. of printed pages: 04

SARDAR PATEL UNIVERSITY
B.C.A. (SEMESTER – II) EXAMINATION
Monday, 6th April 2015
2.30 p.m. to 5.30 p.m.
MATHEMATICS-II (US02FBCA02)

Total Marks: 70

Q.1 Fill in the blanks:

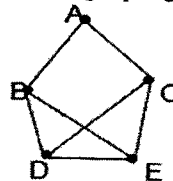
10

1. Graph which contains multiple edges is called _____
 (a) complete (b) Multi-graph (c) simple graph (d) none
2. The degree of isolated vertex is _____
 (a) not possible (b) zero (c) two (d) one
3. The degree of each vertex in complete graph K_n is _____
 (a) $n - 1$ (b) $n + 1$ (c) $2n$ (d) n
4. In a connected map with $V = 5$, $R = 3$ then $E =$ _____
 (a) 2 (b) 6 (c) 4 (d) 3
5. The chromatic number of graph K_{12} is $\chi(K_{12}) =$ _____
 (a) 60 (b) 30 (c) 20 (d) 12
6. If a graph contains a subgraph homeomorphic to $K_{3,3}$ or K_5 then it is a _____ graph.
 (a) complete (b) simple (c) colored (d) non planar
7. $0! + 1! =$ _____
 (a) 0 (b) 1 (c) 2 (d) None of these
8. Quartile deviation is defined as _____
 (a) $\frac{Q_3 - Q_1}{2}$ (b) $\frac{Q_3 - Q_1}{3}$ (c) $\frac{Q_1 - Q_2}{2}$ (d) $\frac{Q_3 + Q_1}{2}$
9. $P(5, 3) =$ _____
 (a) 21 (b) 60 (c) 20 (d) None _____
10. The second quartile Q_2 coincides with _____
 (a) Mean (b) Mode (c) Median (d) Standard deviation

Q.2 Attempt any ten.

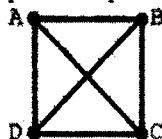
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1. Define with example: Degree of a vertex in a Graph.
2. Draw a diagram of a graph $G=G(V,E)$, where $V=\{A,B,C,D,E\}$,
 $E=[\{A,D\},\{B,B\},\{D,A\},\{B,A\},\{C,A\},\{D,E\}]$
3. Which regular polygon has the same number of diagonals as sides?
4. Find the rest using Euler's formula for the following connected planar multi-graphs having vertices V , edges E and regions R :
 1) $V=10$, $E=14$ 2) $E=6$, $R=3$
5. Find the chromatic number of the graph given bellow:



6. Draw K_5 & $K_{2,3}$

7. Define Planar Graph. Draw the planar representation of the following graph.



8. If the quartiles $Q_1 = 62$, $Q_2 = 142$ and $Q_3 = 195$, then find the Bowley's coefficient of Skewness. Explain the positive correlation with example.

9. Find n , if $P(n, 2) = 72$.

10. Find the number of ways that a party of seven persons can arrange themselves around a circular table.

11. Find standard deviation of the observation 1, 2, 3, 4, 5, 6, 7.

12. Explain the positive correlation with example.

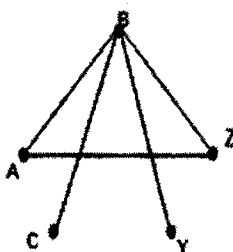
Q.3

(a) Draw the graph G corresponding to each adjacency matrix given below. 05

$$(1) A = \begin{bmatrix} 0 & 1 & 2 & 0 \\ 1 & 1 & 1 & 1 \\ 2 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

$$(2) B = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix}$$

(b) Define bridge and cut points. Find bridges and cut points for the following Graph. 05



OR

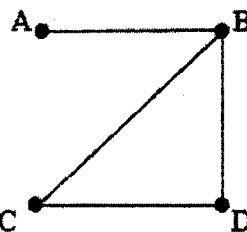
Q.3

(a) Consider the graph:->

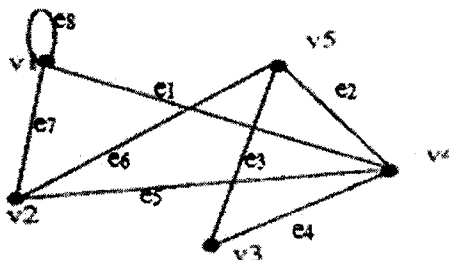
(1) Describe the graph formally.

(2) Find the degree of each vertex.

(3) Verify that the sum of the degrees of the Vertices of a graph is equal to twice the Number of edges.



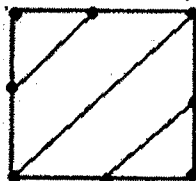
(b) Find the adjacency & Incidence matrix of the graph given below: 05



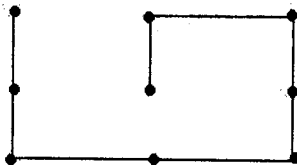
Q.4

- (a) State and Verify Euler's Formula for the following Graphs:

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(1)



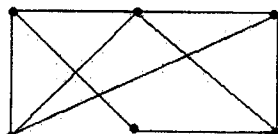
(2)

- (b) Check whether the following graphs are planar or not. Justify your answer.

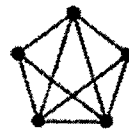
06



(1)



(2)



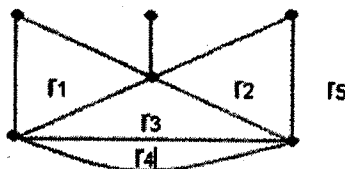
(3)

Q.4

OR

- (a) Identify cycle or closed path that borders each region of the following map. Also find the degree of each region.

04

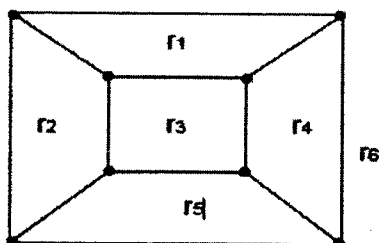


- (b) State Welch-Powell algorithm for painting a graph and give one example of it.

03

- (c) Paint the following maps with minimum number of colors.

03



Q.5

- (a) Find the number n of distinct permutations that can be formed from all the letters of the following words

04

1. MISSISSIPPI
2. THAT
3. BASEBALL
4. NUMERICAL

- (b) Find the number of five letter words which contain three different consonants and two different vowels.

03

- (c) Find n , if $2P(n, 2) + 50 = P(2n, 2)$.

03

Q.5**OR**

- (a) Prove that: $\binom{15}{6} = \binom{14}{5} + \binom{14}{6}$ 05
- (b) Suppose repetitions are not permitted. (a) Find the number of three-digit numbers that can be formed from the six digits 2, 3, 5, 6, 7, and 9. (b) How many of them are less than 400? 05
- (c) How many of them are even?

Q.6

- (a) Find the standard deviation for the following frequency distribution. 06

x:	6	7	8	9	10	11	12
f:	3	6	9	13	8	5	4

- (b) Find Karl Pearson's coefficient of skewness for the following distributions: 04

	Distribution A	Distribution B
Mean	100	90
Median	90	80
S.D.	10	10

OR**Q.6**

- (a) Frequency distribution of the blood pressure given below, Compute the quartile Deviation of the frequency distribution. 06

Systolic BP (mm Hg)	10-20	20-30	30-40	40-50	50-60	60-70	70-80	80-90
No. of Infants	1	6	14	43	21	13	10	1

- (b) Given $\sum X = 125$, $\sum Y = 100$, $\sum X^2 = 650$, $\sum Y^2 = 436$, $\sum XY = 520$ and $n = 25$. Obtain the value of Karl Pearson's coefficient of correlation r . 04
